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Practical 7

* Aim:- Implement a stack using arrays in C++, and perform push, pop, and display operations.

* Theory:- A stack is a linear data structure that follows the LIFO (Last in, First out) principle. This means the element inserted last is removed first, just like stacking plates - only the top plate can be taken off. The size of the stack is fixed using a constant (MAX=5). An integer variable top is used to keep track of the topmost element of the stack. Initially, top is set to -1, which indicates that the stack is empty.

Basic Stack Operations:-

① Push Operation:-

ⓐ The push operation is used to insert an element into the stack.

ⓑ Before inserting, the program checks whether the stack is full.

ⓒ If top == MAX-1, it means the stack is full and stack overflow occurs.

ⓓ Otherwise, top is incremented and the value is stored at stack[top].

② Pop operation:-

ⓐ The pop operation removes the topmost element from the stack.

ⓑ Before removing, the program checks whether

the stack is empty.
(c) If top == -1, stack underflow occurs.
(d) Otherwise, the element at stack[top] is displayed and top is decremented.

(3) Display Operation:-
(a) This operation displays all elements of the stack from top to bottom.
(b) If the stack is empty (top == -1), an appropriate message is shown.
(c) Otherwise, elements are printed using a loop from top to 0.

* Menu-Driven Program
The program uses a menu-driven approach that allows the user to:-
(i) Push element into the stack.
(ii) Pop elements from the stack.
(iii) Display stack elements.
(iv) Exit the program.
This makes the program interactive and easy to understand.

* Conclusion:- In this practical, a stack was implemented using an array in C++. The basic operations such as push, pop and display were performed successfully by following the LIFO principle. The program also handled stack overflow and underflow conditions properly